Victor-Alexandru Pădurean | Curriculum Vitae

Ph.D. Candidate | Helping Humans and AI Work Better Together

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Profile

Ph.D. candidate at the Max Planck Institute for Software Systems researching how humans and AI can work better together. I build and evaluate GenAI systems spanning personalization, model adaptation, agentic pipelines, automated evaluation, and interactive AI tools, combining technical development with large-scale human studies.

Technical Skills

ML / GenAI: PyTorch, Hugging Face Transformers, supervised fine-tuning (SFT), LoRA / PEFT, RAG, LLM agents, multimodal AI, LangChain, LangGraph, LightRAG

Programming / Data: Python, NumPy, Pandas, SQL, Java, C, JavaScript

Systems / Tools: AWS, Docker, Linux, Git, Google Cloud, PostgreSQL, TensorFlow / Keras

Research & Industry Experience

Amazon, Research Scientist Intern — Milan, Italy — Sep 2025 – Jan 2026

- Designed and implemented a knowledge-curation pipeline for AI systems integrating graph-based RAG, code ingestion, multimodal processing, and live retrieval.
- Evaluated retrieval strategies across curated benchmarks and modalities to inform architecture choices for production-oriented deployment.
- Built components using AWS Bedrock, LangChain, LightRAG, and SCIP.

Max Planck Institute for Software Systems, Doctoral Researcher — Saarbrücken, Germany — Jun 2023 – Present

- Build and evaluate GenAI systems for human-AI interaction, spanning fine-tuning, agentic generation, automated verification, and interactive AI tools.
- Research published at NeurIPS, TMLR, ICSE, SIGCSE, ITiCSE, EDM, AIED, LAK, ICER.

Linnify, Data Scientist — Romania — Oct 2020 – Nov 2022

- Researched machine-learning approaches for correlating workplace behavioral data with stress indicators and contributed to early-stage experimentation.

Linnify, Backend Developer Intern — Romania — Jul 2020 – Aug 2020

- Developed and deployed an internal project-management tool using Python, Django, PostgreSQL, Docker, Bitbucket Pipelines, and Google Cloud.

Montran Corporation, Full Stack Developer Intern — Romania — Jul 2019 – Aug 2019

- Built a banking web application using Java, Spring, MySQL, and MongoDB.

Selected Research Projects

UserAlign — NeurIPS 2025

- Developed an inference-time personalization method using best-arm identification in logistic bandits to identify preferred text and image outputs from few pairwise comparisons.
- Evaluated with 960 human participants across text and image generation, outperforming baselines at matched interaction budgets.

Benchmark-CT / LlamaCT — NeurIPS D&B 2024

- Curated a multimodal benchmark comparing generative models with school students on computational reasoning in visual programming.
- Generated 100k+ symbolic training examples and LoRA-fine-tuned Llama3-8B, improving overall accuracy from 22.9% to 53.0%.

Prompt Programming Platform — ITiCSE 2025–26; ICER 2026

- Built a public GenAI programming platform supporting multi-turn dialogue, code editing, execution, and automated testing.
- Deployed across three university cohorts with 2,100+ students to study prompting, code editing, debugging, verification, and voice interaction.

BugSpotter Tool — SIGCSE 2025

- Built an LLM pipeline that generates debugging exercises from specifications and automatically verifies synthesized bugs using executable test suites.
- Deployed with 741 CS1 students; generated exercises varied in difficulty and achieved comparable student performance to instructor-created exercises.

Other Selected Publications

[1] Ruchit Rawal, **Victor-Alexandru Pădurean**, Sven Apel, Adish Singla, and Mariya Toneva. **Hints Help Finding and Fixing Bugs Differently in Python and Text-Based Program Representations.** *ICSE*, 2025.

[2] **Victor-Alexandru Pădurean**, Tung Phung, Nachiket Kotalwar, Michael Liut, Juho Leinonen, Paul Denny, and Adish Singla. **Humanizing Automated Programming Feedback: Fine-Tuning Generative Models with Student-Written Feedback.** *EDM*, 2025.

[3] Manh Hung Nguyen, **Victor-Alexandru Pădurean**, Alkis Gotovos, Sebastian Tschiatschek, and Adish Singla. **Synthesizing High-Quality Programming Tasks with LLM-Based Expert and Student Agents.** *AIED*, 2025.

[4] **Victor-Alexandru Pădurean**, Georgios Tzannetos, and Adish Singla. **Neural Task Synthesis for Visual Programming.** *TMLR*, 2024.

Full publication list on my Google Scholar profile

Education

Max Planck Institute for Software Systems / Saarland University, Germany — Jun 2023 – Present
Ph.D. candidate in Computer Science, Machine Teaching Group
Advisor: Prof. Adish Singla

CS@max planck Program, Germany — Sep 2021 – Jun 2023
Preparatory / course phase

Technical University of Cluj-Napoca, Romania — Oct 2017 – Jun 2021
B.Sc. in Computer Science, GPA 10.0/10.0
Valedictorian, Computer Science English Study Program

Mentorship, Teaching & Service

Mentorship

Jyotika Mahapatra — Aug 2025 – Jul 2026
Master's thesis: *Evaluating the Quality of Tasks in Conversational Programming*

Yang Li — Aug 2023 – Mar 2024
Master's thesis: *Reward Learning for Visual Task Synthesis*

Othman Shbeir — Dec 2024 – Present
Mentorship for Relief Program

Teaching

Generative AI, Teaching Assistant, Saarland University — Winter 2024/25
Reinforcement Learning, Teaching Assistant, Saarland University — Winter 2022/23
Data Networks, Teaching Assistant, Saarland University — Summer 2022

Service

Reviewer for NeurIPS 2025, TMLR 2025, ITiCSE 2025, ISSEP 2025, EDM 2026, Koli Calling 2026, and ACM TOCE 2026.